2025

ICT Lab Assignment #1

Section 3 Task 5

Name: Muhammad Farhan

Roll no: 25K-0538

Program: BCS-1K

Section: 1K

ICT Assignment 1, Section 3, Task 5

Course Instructor: Miss Syeda Tehreem Gilani

Section 3, Task 5: Number System Conversion and Arithmetic

Solve the following problems and present your solutions neatly in a Word document:

- Convert the following numbers n decimal to binary, octal, and hexadecimal: 156, 1024, 255.
- Convert the following into Octal (124)10 , (A78E)16
- Convert the following into Hexadecimal (784)10 , (372)8.
- Convert the following into Binary (235)8 ,(276)10 , (C13E)16 .

Dated:

Name: Muhammad Farhan Roll no: 25k-0538 Program: BCS-1k

Section: k ICT Assignment 1, Section 3, Task 5.

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- Convert the following numbers from decimal to Binary, Octal and Hexadecimal:

i- 156

$(156)_{10}$

Binary: $(156)_{10} \rightarrow (10011100)_2$

Octal: $(234)_8$

Hexadecimal: $(9C)_{16}$

Octal

010 011 100

$2^2 2^1 2^0$ $2^2 2^1 2^0$ $2^2 2^1 2^0$

$0+2+0$ $0+2+1$ $4+0+0$

2 3 4

$\rightarrow (234)_8$

Hexadecimal

1001 1100

$2^3 2^2 2^1 2^0$ $2^3 2^2 2^1 2^0$

$+ +0+1$ $8+4+0+0$

9 $12 \rightarrow C$ $\Rightarrow (9C)_{16}$

Binary Conversion

2	156
2	78 – 0
2	39 – 0
2	19 – 1
2	9 – 1
2	4 – 1
2	2 – 0
	1 – 0 ↑

$(10011100)_2$

ii- 1024

$(1024)_{10}$

Binary = (1000000000)

Octal = $(2000)_8$

Hexadecimal = $(400)_{16}$

010 000 000 000

$2^2+2^1+2^0$ $2^2+2^1+2^0$ $2^2+2^1+2^0$ $2^2+2^1+2^0$

2 0 0 0

$\rightarrow (2000)_8$.

0100 0000 0000

$2^2+2^2+2^1+2^0$ $2^2+2^2+2^1+2^0$ $2^3+2^2+2^1+2^0$

4 0 0

$\rightarrow (400)_{16}$

2	1024
2	512 – 0
2	256 – 0
2	128 – 0
2	64 – 0
2	32 – 0
2	16 – 0
2	8 – 0
2	4 – 0
2	2 – 0
	1 – 0 ↑

$(1000\ 000\ 0000)_2$

Dated:

ii- 255

$(255)_{10}$

Binary = $(11\ 111\ 111)_2$.

Octal = $(377)_8$.

Hexadecimal = $(FF)_{16}$.

2	255
2	127-1
2	63-1
2	31-1
2	15-1
2	7-1
2	3-1
	1-1

011 111 111

$2^2+2^1+2^0$ $2^2+2^1+2^0$ $2^2+2^1+2^0$

$0+2+1$ $4+2+1$ $4+2+1$

3 7 7

$(377)_8$

1 1 1 1 1 1 1 1

$2^3+2^2+2^1+2^0$ $2^3+2^2+2^1+2^0$

$8+4+2+1$ $8+4+2+1$ $(111\ 111\ 11)_2$.

F F

$(FF)_{16}$

• Convert the following into octal

i- $(124)_{10}$

$(124)_{10}$

$\rightarrow (1111100)_2$.

2	124
2	62-0
2	31-0
2	15-1
2	7-1
2	3-1
	1-1

001 111 100

$2^2+2^1+2^0$ $2^2+2^1+2^0$ $2^2+2^1+2^0$

1 $4+2+1$ 4

$(174)_8$.

$\therefore (124)_{10} \rightarrow (174)_8$

ii- $(A78E)_{16}$

2	A (10)	2	7	2	8	2	E (14)
2	5-0	2	3-1	2	4-0	2	7-0
2	2-1		1-1	2	2-0	2	3-1
	1-0				1-0		1-1

1010 0111 1000 1110

001 010 011 110 001 110

$2^2+2^1+2^0$ $2^2+2^1+2^0$ $2^2+2^1+2^0$ $2^2+2^1+2^0$ $2^2+2^1+2^0$ $2^2+2^1+2^0$

1 2 $2+1$ $4+2$ 1 $4+2+0$

1 2 3 6 1 6

$(123616)_8$

$\therefore (A78E)_{16} \rightarrow (123616)_8$.

Dated: _____

• Convert the following into Hexadecimal $(784)_{10}$, $(372)_8$

1- $(784)_{10}$

$(784)_{10}$

$\rightarrow (1100010000)_2$

0011 0001 0000

$2^3+2^1+2^0$ $2^3+2^2+2^1+2^0$ $2^3+2^2+2^1+2^0$

2^4+

$2+1$ 1 0

3 1 0

$(310)_{16}$

$\therefore (784)_{10} \rightarrow (310)_{16}$

2	784
2	392 – 0
2	196 – 0
2	98 – 0
2	49 – 0
2	2^4 – 1
2	12 – 0
2	6 – 0
2	3 – 0
	1 – 1

2- $(372)_8$

2	3	2	7	2	2
	1 – 1	2	3 – 1		1 – 0
	(011)		(111)		(010)

$(372)_8 \rightarrow (011\ 111010)_2$

1 1 1 1 1 0 1 0

$2^3+2^2+2^1+2^0$ $2^3+2^2+2^1+2^0$

$8+4+2+1$ $8+2$

$15 \rightarrow F$ $10 \rightarrow A$

$\therefore (372)_8 \rightarrow (EA)_{16}$

• Convert the following into Binary $(235)_8$, $(276)_{10}$, $(C13E)_{16}$

i- $(235)_8$

2	2	2	3	2	5
	1 – 0		1 – 1	2	2 – 1
	$(010)_2$		$(011)_2$		1 – 0
					$(101)_2$

$\therefore (235)_8 \rightarrow (010\ 011\ 101)_2$

$\rightarrow (010011101)_2$

ii- $(276)_{10}$

$\therefore (276)_{10} = (1000\ 10100)$

2	276
2	138 – 0
2	69 – 0
2	34 – 1
2	17 – 0
2	8 – 1
2	4 – 0
2	2 – 0
	1 – 0

$(100010100)_2$

iii- $(C13E)_{16}$

2	C(12)	1	2	3	2	E(14)
2	6 – 0	$(0001)_2$		1 – 1	2	7 – 0
2	3 – 0			$(0011)_2$	2	3 – 1
	1 – 1					1 – 1
	$(1100)_2$					$(1110)_2$

$\rightarrow (1100\ 0001\ 0011\ 1110)_2$

$(C13E)_{16} \rightarrow (1100\ 0001\ 0011\ 1110)_2$.